Q-BRIDGE: Verifiable Policy Queries over Joint Confidence Views

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***Abstract*—Repeated policy comparisons must distinguish whether a candidate has a guaranteed gain from whether any policy could do better. Q-BRIDGE stores outcome means allowed by observational and randomized logs and declared assumptions in a confidence view. A robust gain holds throughout this view; one coverage event supports adaptive queries. For disjoint nonnegative groups, three-step allocation constructs an optimal policy with matching policy-lower and family-upper certificates, checked independently against original constraints. Exact-LP comparisons agree on all 2,160 distinct synthetic family requests. Holding the view fixed, broader policy choice resolves 107 undecided cases into 86 improvements and 21 exclusions; tightening the original three policies recovers none of the 86. The construction is faster than the exact LP at the smallest tested dimension, slower at larger dimensions. On Coat, seven assumption-grid positions admit above-threshold policies, but all five primary radius-2 views have zero capacity. Execution studies support exact-term retention, but not default screening or family preprocessing on the tested catalogue. Separately, fusion's registered fixed-policy information-gain target is not confirmed. These results separate information, policy-choice and execution limits.**

***Index Terms*—policy evaluation, causal data integration, materialized views, sensitivity analysis, verifiable computation**

I. Introduction

On a fixed production-log snapshot and a smaller randomized experiment, a recommendation analyst asks whether a candidate policy improves over a reference, whether to search further, or whether the evidence excludes the requested guarantee. Unrecorded preferences confound production exposure; experimental information may transfer only for action-group averages.

A policy assigns probabilities to actions. Q-BRIDGE stores a *confidence view*: outcome means consistent with the data and fixed confounding and transfer assumptions. A fixed-policy query tests improvement in every allowed case; a family query asks whether any policy can pass. Reusing checked evidence reduces computation without changing this guarantee.

A running example. Figure 1 has one context, four actions and margin $\delta = .001 = 0.1\,\text{pp}$; m_A is success probability under A. The reference is uniform. Q1 uses policy $(.15, .15, .35, .35)$, hence contrast $c = (-.1, -.1, .1, .1)$. Observational information gives the displayed intervals; randomized information transfers only $(m_A + m_C)/2 = .65$ and $(m_B + m_D)/2 = .70$. Substitution yields

$$\Delta_1 = .27 - .2(m_A + m_B) \geq .01.$$

The vector $(.6, .7, .7, .7)$ attains this one-percentage-point worst-case gain in decimal arithmetic. Interval-only and group-only bounds are $-.075$ and $-.13$; neither certifies improvement. These are constructed teaching inputs, not measured effects.

For Q2 (Fig. 1), a saved certificate proves at least $0.6375\,\text{pp}$ gain, already above the $0.1\,\text{pp}$ requested margin; computing the best bound, $1.05\,\text{pp}$, is unnecessary. For Q3, reuse gives $-2.25\,\text{pp}$ and is inconclusive; solving instead proves $0.5\,\text{pp}$. A certificate is numerical evidence that a separate checker validates. Failed reuse triggers computation, not a negative policy conclusion (Section IV).

The family query asks whether *any* policy can do better. The group sums fix the reference payoff at .675, and $m_A \leq .6$ forces $m_C \geq .7$. Choosing C therefore guarantees $.025 = 2.5\,\text{pp}$. Conversely, the feasible vector $(.6, .7, .7, .7)$ gives no policy more than $.7 - .675$. These matching bounds establish optimality of the robust guarantee, not the actual effect. Section IV-E constructs them.

The workflow is therefore: build one view, check proposed policies, and search or exclude the full family when needed. Our contributions connect these tasks; the sensitivity bounds, simultaneous inference, minimax and dual-reuse principles are established [1], [2], [3], [4], [5], [6].

A source-to-proof interface (Sections II–III). Source assumptions and confidence intervals map into exactly checked constraints. One coverage event—all target means lying in the view—supports new policies and margins. Changing the target, snapshot or assumptions requires a new view.

An exact family answer (Section IV). For disjoint nonnegative groups, we find the smallest feasible ceiling on action means, allocate below it, and recover a policy with an optimality certificate. Construction and checking use Python's standard library and the same output contract as a general exact LP.

An execution comparison (Sections V–VI). Caching exact arithmetic and screening candidates are compared with an-

(a) One fixed view, four action means

Action	Allowed mean	Q1 change
A	[0.15, 0.60]	-0.10
B	[0.20, 0.70]	-0.10
C	[0.25, 0.75]	+0.10
D	[0.30, 0.80]	+0.10

Group means: (A + C)/2 = 0.65

(B + D)/2 = 0.70

(b) Fixed-policy queries

Q1 c = (-0.1, -0.1, +0.1, +0.1)
No saved candidate yet
Solve once **+1.000 pp**

Q2 c = (-0.105, -0.09, +0.105, +0.09)
Cached: +0.638 pp > +0.100 pp threshold
Reuse **+0.638 pp**

Q3 c = (-0.05, -0.15, +0.15, +0.05)
Cached: -2.250 pp < +0.100 pp threshold
Solve after failed reuse **+0.500 pp**

(c) Whole policy family **Best robust guarantee: 2.500 pp**
Policy: choose C with probability 1 Policy lower = family upper (checked exactly)

Constructed illustration • uniform reference • pp = percentage points • displayed values rounded

Fig. 1. One view, two query types. Q1 solves, Q2 reuses evidence, and Q3 solves after reuse fails. Contrasts move action probability; bounds are success-rate gains in percentage points (pp). Panel (c) matches policy lower and family upper bounds. Teaching values are rounded; exported certificates retain exact represented inputs.

swers, evidence and solver paths fixed, including checking cost. A real catalogue tests family preprocessing beyond ordinary caching.

Table I separates the three questions and the evidence needed to answer each.

II. The Query and Its Statistical Contract

A. Target, tables, and declarations

Let $X \in \{1, \dots, K\}$ denote a context, $A \in \{1, \dots, L\}$ an action, and $Y \in \{0, 1\}$ an outcome; write $d = KL$. Source O is observational and source R randomized. The unknown target means are $m^O_{xa} = \mathbb{E}_O[Y(a) \mid X = x]$, while $m^R_{xa} = \mathbb{E}_R[Y(a) \mid X = x]$ refers to the experimental population. For specified context weights w, policy π, and reference π_0, the target contrast is

$$\Delta(c) = \sum_{x,a} w_x(\pi_{xa} - \pi_{0,xa}) m^O_{xa} = c^\top m^O. \tag{1}$$

Each policy row and w is a probability distribution. The caller specifies reference and weights; estimated-weight uncertainty is excluded. Queries test whether worst-case gain exceeds δ (usually $.001 = 0.1$ pp); proofs bind c and δ.

Flatten the context–action pairs into coordinates $i = 1, \dots, d$. The view stores per-action intervals $\ell \le m \le u$ and J group constraints $a \le Tm \le b$. Each row of T weights an action group; a_j, b_j bound its weighted mean. In Fig. 1, the two rows average A/C and B/D. Thus a view is a feasible set, not a point prediction or a table of independent intervals.

Within each source, events (x, a, y) are iid; panels require user-level treatment (Section VI-D). Write N^s_{xay} for counts, $N^s_{xa} = \sum_y N^s_{xay}$ and $S^s_{xa} = N^s_{xa1}$. Randomization requires $A \perp Y(a) \mid X$ and known $q_{xa} = P_R(A = a \mid X = x) > 0$ on moment supports. A union bound permits dependence across sources and rows.

Fix contexts, target, source assumptions, moment rows, sensitivities and confidence allocation before inference, optionally using independent design data. Objectives and thresholds may adapt; new snapshots or confidence constructions require another guarantee.

B. Observational information without a fitted logger

Observed success $p_{xa} = P_O(Y = 1 \mid x, a)$ differs from target mean m^O_{xa}. Let U collect unobserved factors and e_O denote exposure probabilities. Consistency, ignorability conditional on (X, U) and latent overlap give the normalized outcome-conditional weight

$$r^*(x, a, y) = \mathbb{E}_O\left[\frac{e_O(a \mid x)}{e_O(a \mid x, U)} \middle| x, a, y\right],$$
$$m^O_{xa} = \mathbb{E}_O[r^* Y \mid x, a]. \tag{2}$$

Declare $1/\Lambda \le r^* \le \Lambda$, $\Lambda \ge 1$. Larger Λ permits more confounding and hence wider uncertainty. This bounds an outcome-conditional average, not propensity odds or every latent weight. No propensity model is supplied; data alone do not justify this sensitivity parameter [7].

For binary outcomes the conditional-weight relaxation has the known exact interval image [1, Eq. (9)]:

$$L_\Lambda(p) = \max\{p/\Lambda, 1 - \Lambda(1 - p)\},$$
$$U_\Lambda(p) = \min\{\Lambda p, 1 - (1 - p)/\Lambda\}. \tag{3}$$

Substitute $r_1 = m/p$ and $r_0 = (1-m)/(1-p)$ for $0 < p < 1$. Exactness is for this relaxation, not all latent causal models [2].

TABLE I
THREE QUESTIONS OVER ONE FIXED VIEW. GUARANTEES CONCERN EVERY ALLOWED MEAN VECTOR; THEIR FAILURE DOES NOT ESTABLISH A NONPOSITIVE ACTUAL EFFECT.

Question	Returned answer	Independently checked evidence and route
Does this policy guarantee gain $> \delta$?	True, false, or unknown	A dual lower bound or feasible counterexample, obtained by reuse or a fixed-objective solve; overlapping groups allowed.
What is the best robust improvement?	Capacity $\mathcal{R}$ and an attaining policy	Matching policy-lower and family-upper certificates. Our direct construction handles disjoint nonnegative groups; a general exact LP also handles overlaps.
Can every such guarantee be ruled out?	Exclusion when a family upper bound is $\leq \delta$	One feasible point and its maximum contrast over all policies. Full optimization is unnecessary if this witness suffices.

Consistency provides additional information. The same table also implies

$$P_O(A = a, Y = 1 \mid x) \leq m^O_{xa} \leq 1 - P_O(A = a, Y = 0 \mid x). \tag{4}$$

Both fusion and OBS must include these joint-count constraints. With exposure .8 and $p = .5$, they give $[.4, .6]$, versus $[.25, .75]$ from $\Lambda = 2$ alone, without fitting propensities. This changes a substantive baseline comparison (Section VI-C).

C. Which randomized information transfers?

An analyst may declare a jacket-type average stable across sources while allowing individual item means to change. Each row j selects one context x_j, an action subset S_j and normalized weights ν_{ja}. In our representation, H contains the exact row-normalizations of the stored T, fixed before inference (Section III-C). The source declaration concerns this same H:

$$\tau_j = (Hm^R)_j, \qquad |(Hm^O)_j - \tau_j| \leq \rho_j. \tag{5}$$

The tolerance ρ_j permits source differences in a normalized group mean, separately from sampling error; zero requires equal group means. Within a two-action group with positive weights, individual means can change by $(h, -\nu_1 h/\nu_2)$ while preserving that average. Thus agreement on a group does not establish agreement for every action.

At zero tolerance, $c = H^\top v$ implies $c^\top m = v^\top \tau$. A null-space component remains variable wherever the box permits movement. Groups can therefore identify some queries while leaving action means unknown. Full cell transport ($H = I_d$, $\rho = 0$) is stronger; Coat includes its direct-evaluation control.

III. BUILDING ONE CONFIDENCE VIEW

A. Observational intervals and an explicit budget

Here a confidence *budget* is permitted failure probability, not sample size. Split $\alpha = \alpha_O + \alpha_R$ between sources. Of α_O, half covers d two-sided Clopper–Pearson (CP) intervals [8] $[p^-_{xa}, p^+_{xa}]$, each at $\alpha_O/(2d)$. The other half covers $2d$ one-sided joint-event bounds in (4), each at $\alpha_O/(4d)$, with denominator $N^O_x = \sum_{a,y} N^O_{xay}$. Denote their lower endpoints g^1_{xa}, g^0_{xa}. Then

$$\begin{aligned} \ell_{xa} &= \max\{L_\Lambda(p^-_{xa}), g^1_{xa}\}, \\ u_{xa} &= \min\{U_\Lambda(p^+_{xa}), 1 - g^0_{xa}\}. \end{aligned} \tag{6}$$

Monotonicity makes propagation conservative. Conditional binomial coverage permits random stratum counts. Empty action cells contribute $[0, 1]$; empty contexts add no joint restriction. The union bound permits interval dependence.

B. Two prespecified randomized constructions

Allocate α_R to one route, chosen before outcomes: first bound individual experimental means and average their intervals, or directly bound the group's score average. The former is used in the synthetic primary comparison; Coat reports both.

Projected cell intervals. Construct simultaneous binomial intervals $[r^-_i, r^+_i]$ for every randomized cell appearing in H, splitting α_R across those cells. Nonnegativity of H gives normalized group intervals $[Hr^-, Hr^+]$. Only these *group* intervals enter (5); individual experimental means are not equated to target means. Selecting the tighter route after seeing data is not covered by either individual budget.

Group scores. Conditional on an event falling in (x_j, S_j), define $q_j(a) = q_{x_j a}/\sum_{b \in S_j} q_{x_j b}$ and $Z_j = \nu_{jA} Y/q_j(A)$. Then $\mathbb{E}[Z_j] = \tau_j$ and $0 \leq Z_j \leq B_j = \max_a \nu_{ja}/q_j(a)$. Its count, sum and squared sum are obtained from N^R_{xay}, since $Y^2 = Y$. If every score multiplier $\nu_{ja}/q_j(a)$ equals one on S_j, then $Z_j = Y$ and we use a binomial interval. Otherwise, for $n_j \geq 2$, the two-sided empirical Bernstein interval [9] has half-width

$$h_j = \sqrt{\frac{2\widehat{\mathrm{Var}}(Z_j)\log(4/\beta_j)}{n_j}} + \frac{7B_j \log(4/\beta_j)}{3(n_j - 1)}. \tag{7}$$

Here $\beta_j = \alpha_R/J$, and the sample variance uses $n_j - 1$. Clip the resulting expectation interval to $[0, 1]$; for insufficient observations use the full range. Overlapping groups share events but remain covered by the same union bound.

C. Preserving the statistical meaning in stored coefficients

The statistical model describes normalized averages; the checker receives stored numbers. These must describe the same problem. It interprets every stored double-precision (binary64) coefficient as an exact rational. Five stored values of .2, for example, sum to $1 + 2^{-54}$, not one. Treating this row as exactly normalized would break the link to coverage.

For each nonzero stored row T_j, define its exact mass $\kappa_j = \sum_i T_{ji}$ and $H_{ji} = T_{ji}/\kappa_j$. This explicitly defines the normalized estimand and gives $T_j m = \kappa_j H_j m$. Zero rows are deterministic zero constraints. Assignment probabilities

likewise have declared semantics: either exact probabilities summing to one or explicitly normalized nonnegative design weights. Normalizing a vector does not by itself establish that it describes the data-generating assignment law.

For a covered normalized interval $[\tau_j^-, \tau_j^+]$ and a normalized transport tolerance ρ_j, store

$$a_j = \operatorname{down}\big[\kappa_j(\tau_j^- - \rho_j)\big], \quad b_j = \operatorname{up}\big[\kappa_j(\tau_j^+ + \rho_j)\big]. \tag{8}$$

Projected intervals use exact $T_j r^-$ and $T_j r^+$, expanded by $\kappa_j \rho_j$. Observational endpoints, complements, score moments and normalization are computed rationally and rounded outward once at the endpoint. Exact empty intervals are refused before rounding. CP quantiles and the entire floating-point Bernstein half-width kernel (arithmetic, logarithms and square roots) remain trusted; neither these routines nor the scientific declarations are formally verified.

Policy encoding has a separate boundary. If an ideal rational policy contrast c^* is stored as c, let $e = c^* - c$. Because $0 \le m_i \le 1$, a checked bracket $[L, U]$ for the stored objective gives the conservative ideal-policy bracket

$$[L + \sum_i \min(e_i, 0),\ U + \sum_i \max(e_i, 0)]. \tag{9}$$

The additions and threshold comparison are exact. A positive stored bound alone need not certify the ideal policy at zero margin. The fixed-query timing study (V-D) checks stored contrasts; family queries (V-B) use exactly normalized probabilities. The Coat catalogue (VI-D) reports both, applying (9) when needed.

D. Coverage and permitted reuse

Choose $\alpha_O + \alpha_R = \alpha$; fusion uses equal halves in our studies. With endpoints mapped as in (8), save

$$\widehat{\mathcal{C}} = \{m : \ell \le m \le u,\ a \le Tm \le b\}. \tag{10}$$

The exact worst-case bound for the encoded objective is $V(c) = \min_{m \in \widehat{\mathcal{C}}} c^\top m$. Assumptions are declared, interval routines are trusted, and optimization certificates are checked exactly; these are distinct guarantees.

Theorem 1 (One event, all objectives): Under the source and sampling conditions in Section II, $P(m^O \in \widehat{\mathcal{C}}) \ge 1 - \alpha$. On that event, every linear objective, including one chosen adaptively from previous answers, satisfies $V(c) \le c^\top m^O$. Any verified lower approximation to $V(c)$ shares this inequality. A method that cannot establish a usable view or bound makes no positive assertion.

Proof. Condition on independent design information. Conditional interval coverage, averaged over stratum counts, and monotone mapping cover the observational box at $1 - \alpha_O$. The prespecified randomized intervals cover Hm^R at $1 - \alpha_R$; transport and outward mapping give the target moment constraints. A union bound yields $P(m^O \in \widehat{\mathcal{C}}) \ge 1 - \alpha$. On this single event, minimizing over the view bounds every $c^\top m^O$ simultaneously, including adaptive choices. Weak duality gives the same conclusion for checked lower bounds. □

Reuse retains the statistical design [3], [4]. View identity binds count digests, confidence allocation, assignment design, radius, tolerances and matrix; protocols retain provenance.

E. When group information cannot improve the bound

Fusion narrows a fixed box but divides its confidence budget. Let $\mathcal{B}_\alpha \subseteq \mathcal{B}_{\alpha/2}$ be full-budget OBS and fusion's observational box, with minima $B_\alpha, B_{\alpha/2}$. For nonempty $\mathcal{C}_F = \mathcal{B}_{\alpha/2} \cap \{a \le Tm \le b\}$, its optimum obeys

$$V_F - B_\alpha = \underbrace{(V_F - B_{\alpha/2})}_{G \ge 0} - \underbrace{(B_\alpha - B_{\alpha/2})}_{P \ge 0}. \tag{11}$$

Fusion improves the full-budget observational bound exactly when its constraint gain G exceeds its budget cost P.

Proposition 1 (A checkable zero-gain condition): Let $F_c = \arg\min_{m \in \mathcal{B}_{\alpha/2}} c^\top m$. Then $G = 0$ if and only if F_c contains a point satisfying all randomized group constraints. If F_c is a singleton, checking its moment rows decides whether $G = 0$ or $G > 0$.

Proof. A feasible box minimizer attains the box bound in $\mathcal{C}_F$. Conversely, compactness of nonempty $\mathcal{C}_F$ gives an optimizer; equality places it in F_c, and disjointness forces a strict gap.□

Choose box endpoints by the signs of c and check the group rows. A feasible point proves zero constraint gain; if zero coefficients leave multiple minimizers, failure of one point is inconclusive. Section VI-C applies this fixed-view diagnostic to actual losses.

IV. Queries with Checkable Answers

Two query modes share the view: test one supplied policy (A–C), or find the best policy and rule out further gains (D–E). A *producer* proposes numerical evidence; an independent *receiver* checks it against the caller's view. For a fixed policy, reuse is tried before a persistent LP. True certifies the requested robust gain; false exhibits an allowed case that fails it; unknown means the available evidence cannot decide.

A. Materialization and lower evidence

For independent binary events, each source's $2d$ counters determine its score moments and, with fixed declarations, its constraints. Aggregation costs $O(n_O + n_R)$ work and $O(d)$ storage; new objectives require no rescan. User panels additionally require group-score variances (Section VI-D).

A usable view requires an exactly checked feasible point, proposed by the solver or slack maximization. Otherwise its status is `uncertain`: no certificate or decision (mathematically unknown). This gate prevents empty-view certification but neither proves emptiness nor reliably accepts zero-volume sets.

The certificate's nonnegative multipliers η_L, η_U weight the lower and upper group constraints. These *dual prices* cancel or bound parts of the objective, leaving a box minimum that is easy to check. Let $z = c + T^\top(\eta_U - \eta_L)$. Weak duality gives

$$D(c; \eta) = a^\top \eta_L - b^\top \eta_U + \sum_i \min\{z_i \ell_i, z_i u_i\} \le V(c). \tag{12}$$

Subtracting lower constraints and adding upper constraints cannot increase a feasible objective. Minimizing over the box proves (12); any nonnegative candidate is valid, with optimality affecting only tightness [5].

The standard-library checker interprets represented coefficients exactly and rounds the final bound downward. Proofs bind objective, threshold and view digest; the caller supplies the trusted view. Digests do not authenticate raw data. Nonfinite or failed proposals cannot authorize success. Independent optimization certificates are established [10].

B. Reusing exact terms

A bounded first-in, first-out (FIFO) bank stores each dual divided by its old objective's binary64 ℓ_1 norm, then uses the new norm as scale s. Zero old norm skips insertion; zero new norm uses $s = 0$ without division. The checker uses the saved base and scale exactly, not an unrounded ideal ratio. Floating scores order candidates; only exact bounds authorize answers. For a base dual, retain $k = a^\top \eta_L - b^\top \eta_U$ and $r = T^\top(\eta_U - \eta_L)$. For a nonnegative represented scale s,

$$D(c; s\eta) = sk + \sum_i \min\{(c_i + sr_i)\ell_i, (c_i + sr_i)u_i\}. \quad (13)$$

Setup costs $O(\mathrm{nnz}(T) + d + J)$ rational operations and later checks $O(d)$, with coefficient-dependent bit cost. Proofs preserve the exact base–scale product; rounding multipliers separately can change last bits. A fresh receiver reconstructs (13); retained state is bound to its view and base and reconstructed or bypassed on validation failure. Dual and bounded-Lagrangian reuse are established [6], [11], [12]; persistent LP reoptimization supplies a control [13].

In the decimal running example, Q1 supplies constraint weights $\eta_L = (.2, .2), \eta_U = 0$, giving $k = .27$ and $r = (-.1, -.1, -.1, -.1)$. Relative to these weights, Q2 scales by .975 and Q3 by 1; substitution in (13) gives .006375 and $-.0225$. We reuse constraint weights, not the feasible point (which supplies an upper bound).

C. Thresholds, screening and fallback

A checked feasible point brackets the optimum [14, Sec. 5.5.1]:

$$L(c) \le V(c) \le U(c) = c^\top \widetilde{m}. \quad (14)$$

Return true for $L > \delta$, false for $U \le \delta$, and unknown otherwise. False exhibits failure permitted by the view, not failure of the actual effect; a nonpositive lower bound alone decides nothing. Artifact status `unresolved` means no positive lower certificate and can accompany either false or unknown; the decision field distinguishes them. View witnesses and previous solver points remain feasible after objective changes. Check global feasibility before retention, round later dot products upward, and export the point bound to its view and objective for independent verification.

The path is: validate the view, check ordered reusable bounds, decide if possible, otherwise call the LP and check its evidence. Candidates start with zero dual, last native dual, initial feasible point and last native point. Histories retain at most 16 candidates of each enabled kind, including the last; reuse hits do not replace native last-result state. Native dual-simplex cutoff proposals [15] undergo exact checking; undecided requests resume without the cutoff and may remain unknown. Checked native duals update the bank; zero-dual answers are not dual reuse.

An objective cache reuses a result for the *same* c; the candidate bank reuses constraint weights for *new* c. Retention caches their exact k, r terms. Screening instead asks cheaply whether a candidate could decide the query, using outward binary64 intervals. It defers a lower candidate when its upper endpoint is $\le \delta$, or a primal candidate when its lower endpoint is $> \delta$. All operations expand outward; overflow or unbounded intervals take the exact path. Screening alone cannot authorize an answer.

True can bypass insufficient deferred lowers. Before false, replay all deferred lowers to preserve the best exported lower; before solving, replay all remaining candidates in order. Evidence, native state and solver paths must match the unscreened plan. Only zero dual before last-dual true, or initial primal before last-primal false, can be permanently omitted. Complete service cost determines whether these omissions pay.

Structural controls use box endpoints, continuous knapsack for single rows [16], or sparse HiGHS for overlaps [17]; only the reconstructed global dual is certified.

D. Certifying a whole policy family

Here $\mathcal{C}$ is the fixed materialized view; stars mark exact probability representations rather than a previously rounded contrast. The family Π contains all randomized policies on the available actions. Its *capacity* is the largest worst-case improvement attainable by any policy. Fix a nonempty compact convex view $\mathcal{C} \subseteq [0,1]^{KL}$, exact probability weights $w^* \ge 0$, $\sum_x w^*_x = 1$, and reference $\pi^*_0 \in \Pi$. Define

$$\mathcal{R}(\mathcal{C}) = \max_{\pi \in \Pi} \min_{m \in \mathcal{C}} \sum_{x,a} w^*_x(\pi_{xa} - \pi^*_{0,xa}) m_{xa}, \quad (15)$$

$$F(m) = \sum_x w^*_x \left[\max_a m_{xa} - \sum_a \pi^*_{0,xa} m_{xa} \right]. \quad (16)$$

The reference belongs to Π, so $\mathcal{R} \ge 0$. At one allowed mean vector, $F(m)$ is the gain of choosing its best action in each context. Every checked $m \in \mathcal{C}$ gives $\mathcal{R}(\mathcal{C}) \le F(m)$: each policy's minimum is at most its payoff at this point, whose policy maximum is $F(m)$. Thus $F(m) \le \delta$ excludes every robust gain strictly above δ. The witness need not equal the true means; it does not bound actual effects.

Proposition 2 (Zero capacity in a fixed view): $\mathcal{R}(\mathcal{C}) = \min_{m \in \mathcal{C}} F(m)$. If all $w^*_x > 0$ and $\pi^*_{0,xa} > 0$, then $\mathcal{R}(\mathcal{C}) = 0$ exactly when $\mathcal{C}$ contains a vector constant across actions within every context.

Proof. Compact convex domains and a continuous bilinear payoff permit minimax exchange [18]; row maximization gives F. Its nonnegative summands, under strict positivity, vanish exactly at constant rows. Compactness attains the minimum. □

Without strict positivity, constant points remain sufficient for zero capacity, but the converse need not hold.

For nonnegative single-context moment rows, even overlapping ones, form

$$I_x = [\max_a \ell_{xa}, \min_a u_{xa}] \cap \bigcap_{j:x_j=x} [a_j/\kappa_j, b_j/\kappa_j].$$

Here $\kappa_j = \sum_i T_{ji} > 0$; zero rows require $a_j \le 0 \le b_j$. Choose one $v_x \in I_x$ per context, set every $m_{xa} = v_x$, and check the original constraints. This *constant-action witness* proves zero capacity without optimizing. Under the proposition's positivity conditions, an empty intersection proves $\mathcal{R} > 0$, not $\mathcal{R} > \delta$ for positive δ.

Partial evidence can bracket the family without full optimization. The maximum of checked lower bounds for any legal policies, together with the reference's zero, is $L_{\text{saved}} \le \mathcal{R}$. A nonempty set M_{saved} of checked points gives $U_{\text{saved}} = \min_{m \in M_{\text{saved}}} F(m) \ge \mathcal{R}$. Share them only for identical $\mathcal{C}, w^*, \pi_0^*$. Then $L_{\text{saved}} > \delta$ certifies existence, $U_{\text{saved}} \le \delta$ excludes it, and otherwise the family remains unresolved. Equation (18) separates certificate slack from restricted policy choice.

E. Exact capacity for partitioned moments

Minimax turns policy search into minimizing $t - p^\top m$, with $m_i \le t$ and reference probabilities $p = \pi^*_{0,x}$ in one context. For disjoint groups, clipping and repairing means permits the smallest feasible ceiling t_0 without increasing this gap. We maximize the reference payoff below t_0, then recover a policy attaining that upper bound. The exported proof uses the *original* constraints, not the artificial ceiling. Minimax and continuous allocation are classical ingredients [19], [20].

Theorem 2 (An attaining policy with an original-view certificate): Let $\mathcal{C}$ be a nonempty rational box-and-moment view with $0 \le \ell \le u \le 1$. Suppose every nonzero moment row is nonnegative, belongs to one context, and has positive support disjoint from every other row. For any rational probability weights w^* and reference policy π_0^*, including zero entries, the three-step construction returns $m^* \in \mathcal{C}$, $\pi^* \in \Pi$ and nonnegative original-row multipliers η_L, η_U such that, with $c^*_{xa} = w^*_x(\pi^*_{xa} - \pi^*_{0,xa})$,

$$D(c^*; \eta) = c^{*\top} m^* = F(m^*) = \mathcal{R}(\mathcal{C}).$$

The dense implementation uses $O(dJ + d^2)$ exact arithmetic/comparison operations; the independent receiver uses $O(dJ + d)$. These are unit-cost rational-operation counts, excluding coefficient-dependent bit costs, encoding and hashing.

The certificate closes a sandwich: $D(c^*; \eta) \le \min_{m \in \mathcal{C}} c^{*\top} m \le \mathcal{R} \le F(m^*)$. Equal checked endpoints prove optimality independently of how the producer found them. Check the domain before construction; reject unsupported rows. Zero rows require $a_j \le 0 \le b_j$ and receive zero multipliers. Ties use index order. The following three steps establish the theorem.

Step 1: Find the smallest feasible cap.

Within this algorithm, $p = \pi^*_{0,x}$ is the reference distribution, distinct from observed success p_{xa}. Write $\ell \le m \le u$ and $a_j \le T_j m \le b_j$. Set

$$t_\ell = \max_i \ell_i, \qquad g_j(t) = \sum_i T_{ji} \min(u_i, t),$$
$$t_0 = \min\{t \ge t_\ell : g_j(t) \ge a_j \quad \forall j\},$$
$$h_i = \min(u_i, t_0).$$

Here j ranges over nonzero rows. Find lower-bound attainment along the rational upper-box breakpoints: each row sum is piecewise linear, and the largest required root, or t_ℓ, determines the cap. Only lower moments force the ceiling upward. Nonemptiness gives $T_j \ell \le b_j$; each disjoint group can independently choose a mass in $[\max(a_j, T_j\ell), \min(b_j, T_j h)]$. Thus $\mathcal{C} \cap [\ell, h]$ is nonempty, although not every point of the box is feasible. Every point in this intersection has maximum t_0, by the cap's minimality.

Minimum-cap sufficiency. For any feasible m with $t = \max_i m_i$, clip to $v_i = \min(m_i, t_0)$. This preserves lower boxes and upper moments. Repair violated lower moments within $[v, h]$ up to $a_j \le b_j$; the cap definition permits each repair, and disjoint supports prevent interference. No repaired row exceeds its upper bound, and untouched rows remain feasible. The resulting $v' \ge v$ is feasible. Since p is a probability vector, $p^\top(m - v) \le t - t_0$, giving

$$\max_i v'_i - p^\top v' \le t_0 - p^\top v \le t - p^\top m.$$

Thus maximizing $p^\top m$ within this cap minimizes the original gap.

Step 2: Allocate means within the cap. This step maximizes $p^\top m$ under the cap and original moments. Increasing m_i by ϵ consumes $T_{ji}\epsilon$ of group budget and gains $p_i\epsilon$ of reference payoff. Starting at ℓ, fill each nonzero row by decreasing this benefit/cost ratio p_i/T_{ji}, only on $T_{ji} > 0$, to total mass $\min(b_j, T_j h)$; uncovered coordinates take h_i. The additional mass is $\min(b_j, T_j h) - T_j \ell$. Nonnegative reference weights make this fill maximize the reference payoff within the cap, even when zero-ratio coordinates receive the remaining mass. Its total meets the original lower bound. For its nonnegative upper price λ_j, first use zero if $b_j \ge T_j h$, including equality. Otherwise use the largest support ratio when $b_j - T_j \ell = 0$; for positive budget use the last ratio receiving strictly positive mass, including exact coordinate-boundary exhaustion. These rules include fixed coordinates and zero reference probabilities. Writing $r = p - T^\top \lambda$, the allocation satisfies

$$r_i > 0 \Rightarrow m^*_i = h_i, \quad r_i < 0 \Rightarrow m^*_i = \ell_i,$$
$$\lambda_j(b_j - T_j m^*) = 0.$$

Step 3: Recover the policy and original-row prices. We need a probability vector supported on maximizers of m^* and prices making the original-box bound (12) tight there. The initial policy cancels positive residuals at capped coordinates; a tight original constraint supplies missing probability mass. Initialize

$$\widetilde{\pi}_i = (r_i)_+ \mathbf{1}\{u_i \ge t_0\}, \qquad D_\pi = 1 - \sum_i \widetilde{\pi}_i \ge 0.$$

Here $0 \le \widetilde{\pi}_i \le p_i$, with positive mass only at maximum coordinates. The original-box coefficient $z_i^{\text{init}} = \widetilde{\pi}_i - r_i$ is positive only at ℓ_i and negative only at u_i: when $u_i < t_0$, $h_i = u_i$; otherwise initialized policy mass cancels positive reference residuals.

If $t_0 = t_\ell$, add D_π to a coordinate with $\ell_i = t_0$ and set all lower prices η to zero; its box coefficient increases at this lower endpoint. Otherwise choose the first row attaining its lower bound at t_0. Since $T_j h = a_j$, its supported means must equal h_i, and zero-price priority gives $\lambda_j = 0$, even if its upper moment is also tight. Set

$$E = \{i : T_{ji} > 0,\ u_i \ge t_0\}, \qquad s = \sum_{i \in E} T_{ji} > 0,$$

where positivity follows because otherwise this row would attain its bound below t_0. Set $\eta_j = D_\pi / s$, with other lower prices zero. Add $\eta_j T_{ji}$ to $\widetilde{\pi}_i$ on E. Updated box coefficients are zero on E and $-p_i - \eta_j T_{ji} \le 0$ elsewhere in this row, where $m_i^* = h_i = u_i$; other coefficients are unchanged. The non-strict $u_i \ge t_0$ includes roots at breakpoints.

The completed π sums to one and selects maximum coordinates. With $z = \pi - p + T^\top(\lambda - \eta)$, box signs and tight priced rows make (12) exact for $c = \pi - p$ and $(\eta_L, \eta_U) = (\eta, \lambda)$:

$$\begin{aligned} D(\pi - p; (\eta, \lambda)) &= (\pi - p)^\top m^* \\ &= t_0 - p^\top m^*. \end{aligned}$$

Scale each context's objective and both row-price types by the same w_x^*. Concatenation yields original-program lower $= F(m^*) = \mathcal{R}(\mathcal{C})$, including zero-weight contexts. Such contexts still require a feasible point and a legal policy; only their objective and multipliers vanish.

The receiver checks original constraints, legal policy probabilities, multiplier signs and lower/objective/upper equality against the caller's request. It trusts neither the artificial cap nor the producer's internal state. □

The running example, step by step. Using Fig. 1's decimal inputs, Step 1 gives $t_0 = .7$. Step 2 gives $m^* = (.6, .7, .7, .7)$ and $\lambda = (0, 0)$. Step 3 starts with $(0, .25, .25, .25)$, adds the missing .25 to C, and returns $\pi^* = (0, .25, .5, .25)$ with $\eta_L = (.5, 0)$ and $\eta_U = 0$. The lower bound $.65(.5) - .6(.5) = .025$ equals the family upper bound $.7 - .675 = .025$. Pure C in Fig. 1 is another optimal policy. This exact decimal illustration explains the construction; measured runs check their own represented inputs.

V. Evaluation Design

The experiments ask four distinct questions. The registered fixed-policy comparison tests whether combining sources supplies more information (VI-C). Retrospective family queries test completeness and the value of broader policy choice (VI-A); controlled execution measures arithmetic reuse (VI-B). Coat tests a real policy decision and reuse on an unchanged catalogue (VI-D). These are different claims, not independent replications of one success rate. Complete-solver and new Coat timings use an AMD EPYC 7763, Python 3.12.3 and one CPU/native thread.

A. Fresh worlds and policy design

The registered study draws 48 confounded worlds: $(K, L) \in \{(2,4), (8,8), (16,16)\}$, uniform/skewed randomization and group drift $0/.02$, with four worlds per stratum and five inference samples per world. Twelve additional worlds test near-unit sensitivity. A latent binary factor affects exposure and rewards. Metadata $h_a \in [-.5, .5]$ are fixed without inference labels, but their designed positive reward association limits representativeness. Random groups allow individual shifts while controlling group drift; shifts are not optimized for the evaluation objective.

Inference uses $2{,}000d$ O events and $100d$ or $1{,}000d$ R events. Independent design data use $500d$ O and $100d$ R events. Against a uniform reference, the three policies are: $\mathrm{softmax}(1.5h)$; $\mathrm{softmax}(4\widehat{p}_x)$ with design-R $\widehat{p}_{xa} = (S_{xa} + .5)/(N_{xa} + 1)$; and the design-view choice with the largest checked positive lower bound among twelve tilts. The tilts use linear or alternating action scores and slopes $\{-2, -1, -.5, .5, 1, 2\}$; ties retain the first positive maximum, and otherwise the reference. Design outcomes never enter inference counts. We compare $\Lambda = 2$ with an oracle boundary computed from simulator truth and rounded upward. This diagnostic preserves counts and policy rules, but the design-selected policy can vary with radius; it does not isolate a radius effect or replace the registered comparison.

The primary endpoint compares projected fusion with certification by either full-α OBS or projected R, for the metadata policy, low R budget and $\Lambda = 2$. Average paired certification differences within each of 48 independent worlds. Their values lie in $[-1, 1]$, so the one-sided 95% Hoeffding lower bound is the mean minus $\sqrt{2\log(20)/48}$. The registered target requires this endpoint > 0.05. Other settings and the stratified t interval are descriptive; the standalone union is a comparator, not a jointly calibrated decision rule.

B. Whole-family queries on fixed views

The retrospective family study reuses the 48 main worlds and the stored threshold for .001. Its 24 comparison cells are three source constructions (OBS, projected R, projected Fusion) × two R sample budgets × two radii × finite/population information. Each cell retains 48 worlds × five repetitions, giving 5,760 comparison positions, not independent datasets. The underlying 2,880 finite and 288 population program records reduce to 2,160 distinct exact $(\mathcal{C}, w, \pi_0)$ requests. Both solvers run each request: 4,320 synthetic calls. Deduplication saves computation without dropping comparison positions.

We decode saved inputs exactly, normalize policy probabilities rationally and recheck each position's three saved duals. Among 240 primary finite-Fusion cases, 12 already have checked policy gains; 121 have checked witnesses constant across actions within each context, with $F(m) = 0$, constructed before complete family optimization. The remaining 107 are the fixed recovery set. This family-level uncertainty differs from a fixed policy lacking a positive lower bound. Theorem 1 permits selection on the inference view; this

retrospective analysis is not a new confirmatory or out-of-sample learning test.

C. A complete exact solver and the source of recovery

SoPlex 8.1.0 in rational mode [21], [22], [23] solves the epigraph LP on the identical $(\mathcal{C}, w, \pi_0)$:

$$\min_{m\in\mathcal{C},t} \sum_x w_x\big(t_x - \pi_{0,x}^\top m_x\big), \qquad m_{xa} \le t_x, \quad t_x \text{ free.} \tag{17}$$

Its epigraph multipliers $\mu_{xa} \ge 0$ sum to w_x in each context. Set $\pi_{xa} = \mu_{xa}/w_x$ when $w_x > 0$, and use the reference row when $w_x = 0$. Original-row duals and a feasible point complete the same certificate as the partition construction. Exact import/export and independent checking are required; solver status is insufficient. Neither the construction's answer nor a saved point is supplied.

Two lowest hashes per size cell select 78 unique requests (72 cells, 144 memberships) for three paired rounds in alternating order. Service runs from post-initialization transmission through checked certificates, including construction, verification, serialization and transport; startup, diagnostics and batch totals are separate. Fresh models/workers use one CPU, ten-second service and 8 GiB address-space limits; all phase limits are archived. Failures retain costs and denominators; ratios require complete pairs. Rounds measure machine variation, not independent samples.

Exact optima L_i^* for each occurrence's three normalized contrasts distinguish certificate transfer from policy choice: 7,762 unique $(\mathcal{C}, c)$ requests and 17,280 policy slots. Set $L_3^+ = \max(0, L_1^*, L_2^*, L_3^*)$; zero is the reference policy. With the unchanged saved lower rule L_{saved},

$$\mathcal{R} - L_{\text{saved}} = \underbrace{L_3^+ - L_{\text{saved}}}_{\text{certificate-transfer gap}} + \underbrace{\mathcal{R} - L_3^+}_{\text{three-policy restriction}} . \tag{18}$$

Both terms are nonnegative. Saved duals solve their stored objectives, but exact policy normalization can loosen transferred bounds. Missing optima leave the decomposition unknown. The 107-case recovery set and threshold remain unchanged.

Natural metadata partitions. Coat gender, jacket type and color each partition 300 items. Separate row families preserve the original 31-row views' exact coefficients, endpoints and confidence allocation. The evaluation includes 144 natural-row subviews, twelve box-only OBS positions, the 60-view grid and one full-cell control: 217 memberships, 125 unique requests. The construction supports 95; the generic LP covers 30 overlapping-row requests too. Subviews relax their parents and cannot increase capacity. Coverage uses each parent's event, without a joint 95% guarantee over all memberships. A feasible common point may exclude every policy; this is not exclusively a family-optimization benefit.

D. Repeated fixed-policy queries

The 63 inputs are three primary 1,000-query streams sharing one connected 512-cell view, 54 geometry streams ($d = 64, 512, 2048$; disjoint/mixed/connected rows), and six source-derived streams. The latter use generated logs with 64 actions: one million O and four million R events, with uniform/skewed assignment. Broader streams have 100 queries each. Nearby queries perturb contrasts; unrelated directions preserve legal policies; $\delta = .001$ throughout. Primary streams emphasize reuse: one solve, 999 hits, all true; broader streams permit all three verdicts. These constructed workloads do not estimate online hit rates; source ingestion is excluded.

Four plans cross interval screening S with cross-query exact-term retention P, sharing native configuration, zero-dual/last-dual/primal order, proof format and receiver. $P = 0$ discards exactly the dual constants and residuals retained by $P = 1$. Optimized plans equally retain view decoding, primal state and within-request objective/box conversion; sparse residuals skip zeros. The nonfactorial last-first bridge uses S_0P_1 arithmetic, checking last before zero while preserving decisions and native opportunities; valid true certificates may differ.

Joint service includes fresh producer/receiver initialization, view validation, queries, failures, native fallback, serialization and persistent FIFO16 receiver verification; disk I/O, compression, network transfer and separate legacy/Fraction audits are excluded. Core plans must match decisions, evidence and native paths.

All inputs undergo one quality pass and five timing rounds (45,000/225,000 responses), with fresh arm-streams, one CPU/native thread and input-independent shuffled order. Equal-weight geometric means within each input group/round retain four conditional factor ratios, their log interaction and four bridge ratios. Input ratios, slow cases, absolute costs, receiver shares and prefixes are archived. Registered checks are descriptive: retention helps without screening; screening helps without retention but costs with retention. Five rounds measure machine variation, not independent workloads.

VI. Results and Boundaries

A. What complete family optimization adds

Both methods agree on all 2,160 unique synthetic requests (5,760 memberships). These 4,320 responses, 7,762 fixed-policy optima, 220 Coat calls and 468 timing calls all pass independent verification: 12,770/12,770, without timeout or incomplete call (Fig. 2).

Broader policy search changes the fixed-view answer. The 107 previously unresolved primary finite Fusion cases split into 86 above-threshold and 21 at-most-threshold family answers. Exact reoptimization of the same three policies recovers none: all 86 gains require different policies, rather than tighter transferred certificates. The 21 exclusions have $0 < \mathcal{R} \le \delta$, so positivity alone would not justify continuing search. Together with the earlier 12 above-threshold and 121 zero-capacity cases, the totals are 98 and 142 out of 240. Across all 86 recovered cases, the attainable gain has median 0.991 pp, interquartile range $[0.583, 1.549]$ pp and range $[0.105, 3.632]$ pp. For every one, the three-policy optimum after including the reference is zero. These are robust guarantees under the same view, not measured effects or evidence that adding a data source improves information.

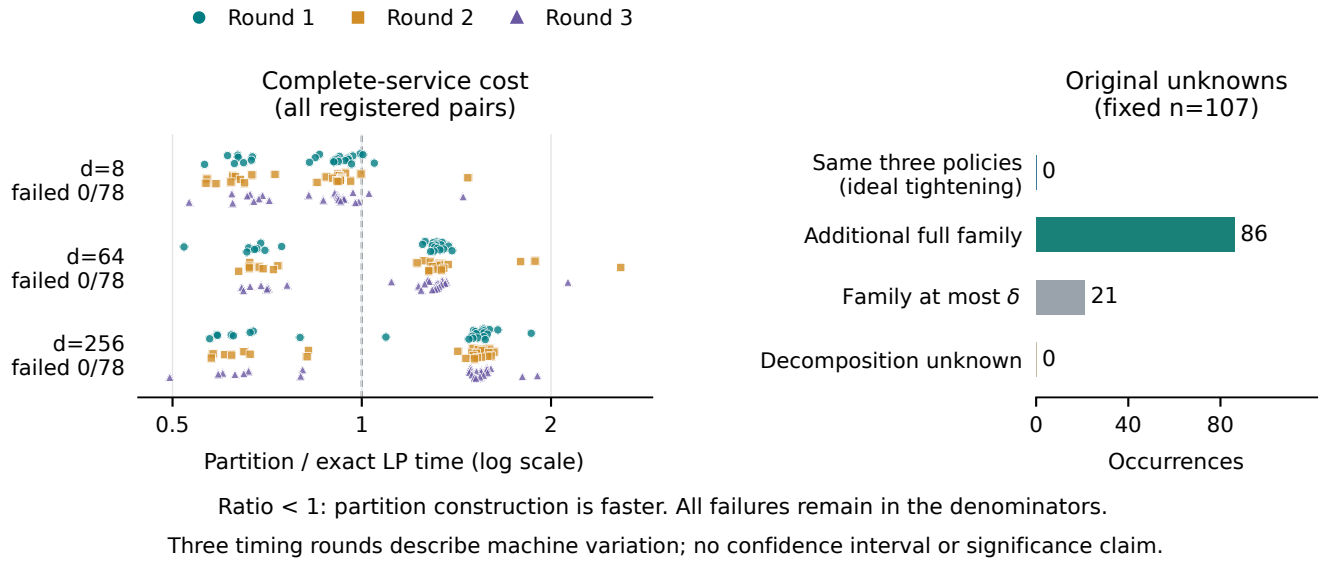


Fig. 2. Checked cost and fixed-view recovery. Left: all 234 pairs (78 inputs, three rounds), after worker readiness through verification; startup/cleanup are separate. SoPlex uses the same receiver; offsets separate points. Right: all 107 undecided queries. Tightening rechecks existing policies at ideal objectives; additional family permits different policies. No failures or missing decompositions; rounds describe machine variation, not confidence intervals.

TABLE II
MEDIANS OVER 78 CALLS PER METHOD/DIMENSION. RESPONSE: AFTER WORKER INITIALIZATION THROUGH PROOF CHECKING; CHECK: PER-CALL RECEIVER SHARE; WHOLE: INCLUDES STARTUP/CLEANUP. PART: PARTITION CONSTRUCTION; LP: SOPLEX. KB: 1,000 BYTES.

d	Method	Response ms	Check %	Whole ms	kB
8	Part	4.66	24.4	122.2	3.47
8	LP	4.99	22.1	135.2	3.47
64	Part	53.58	23.2	172.2	20.93
64	LP	40.85	31.6	170.0	20.91
256	Part	648.13	22.8	770.3	149.51
256	LP	419.97	34.6	554.0	149.44

Equal answers do not imply uniform speed gains. All 234 timing pairs succeed. Across three rounds, the construction/SoPlex geometric-mean service ratios are 0.808–0.835 at $d = 8$, 1.066–1.129 at $d = 64$, and 1.170–1.196 at $d = 256$. The construction is slower in 5/78, 55/78 and 54/78 pairs, respectively. These compare checked service for a Python rational construction and compiled rational LP, without establishing asymptotic advantage. All pair costs, certificate sizes, memory observations and failure denominators are archived. Table II reports absolute costs. At $d = 256$, verification alone takes median 144.3/144.6 ms (Part/LP), with about 149.5 kB per certificate. These pooled medians describe evidence cost, not matched-call ratios.

Better selection does not guarantee a better view. At the larger conservative budget, projected R passes 223/240 thresholds versus Fusion's 218/240; R has greater capacity in 128 pairs, Fusion in 100, with 12 ties. At the smaller oracle-boundary budget, OBS and Fusion both pass 239 thresholds, yet OBS has greater capacity in 219 pairs and Fusion in 21. Their full- and split-budget views are not nested. All 24 cells and exact decompositions remain available. In the partition study's 2,880 finite views, the generating means satisfy the view and every returned-policy assertion is correct against the stored truth. These retrospective checks do not establish uniform coverage or validate the source assumptions.

B. The cost of answering repeated policy queries

All 270,000 quality and timing responses pass the recorded independent exact replay; the four core plans agree on evidence, decisions and native paths. Figure 3 shows the five optimized rounds. On the three primary streams, retention's joint-cost ratios are 0.735–0.748 without screening and 0.766–0.791 with it. Screening costs 1.103–1.138 and 1.165–1.186 times the corresponding unscreened plans. It also loses in every engine and source-derived group-round mean. Its earlier benefit without retention therefore does not survive optimized arithmetic; only two of the three registered directions hold in every primary round.

The last-first bridge costs 0.833–0.882 times S_0P_1 on primary streams, but is slower in one of their 15 input-round pairs. The corresponding slower counts are 66/270 engine and 17/30 source-derived pairs. Retention also has adverse individual cases despite lower group means. These results support retaining reusable exact terms, not default screening or universal speedups; the bridge's changed candidate order is a separate mechanism.

a) Auxiliary execution controls: Under fixed arithmetic, screening plus retention fails the registered joint-cost-ratio target ≤ 0.8 against each feature alone. Screening slows every broader-group round. A larger bank misses its speed target; changed decisions preclude isolated reuse attribution. Materialization helps ingestion, but preparation costs 1.580/1.619

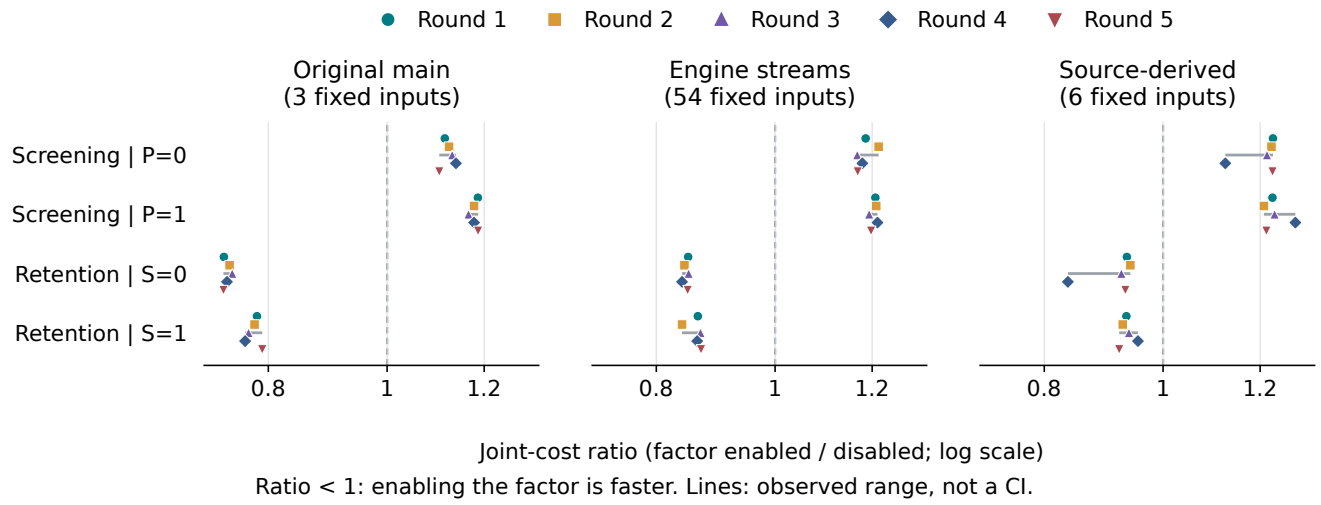


Fig. 3. Retention lowers group-average cost; screening does not. S: screening; P: cross-query retention. Dots: each round's equal-input geometric-mean cost ratio (below one favors enabling the factor). Lines: five-round range, not confidence intervals. All three input groups and four conditional comparisons appear; interaction and last-first comparisons are archived.

TABLE III
CONSERVATIVE, LOWER-BUDGET DIAGNOSIS; 240 QUERIES PER POLICY. FINITE: THE FINITE OPTIMUM FAILS BUT THE POPULATION OPTIMUM PASSES. POP.: THE TRUE EFFECT EXCEEDS δ BUT BOTH OPTIMA FAIL. NO QUERY IS MISSED SOLELY BY NUMERICAL OPTIMIZATION.

Policy	Cert.	Finite	Pop.	True $\leq \delta$
Metadata	12	33	195	0
R heuristic	4	43	190	3
Robust design	2	0	0	238

times hot execution under uniform/skewed assignment, respectively. Full controls remain in the artifact.

C. What limits information from fusion

All 60 worlds and 1,200 inference configurations complete. Primary fusion certifies 12/240 versus six for the standalone union (OBS six, R zero): six additions, no losses, mean gain 0.025. Hoeffding's lower endpoint -0.3283 misses the registered target; its large finite-world penalty limits power for modest gains. The descriptive t interval is $[-0.0249, 0.0749]$. All additions occur in two zero-drift 2×4 worlds. This does not confirm the target or establish absence of benefit.

Larger-budget projected fusion certifies 15 metadata queries, 16 heuristic queries and two robust-design selections. More randomized traffic does not resolve transfer or radius uncertainty.

Sampling uncertainty versus ambiguity. Finite views include sampling intervals; population views use exact generating statistics, retaining ambiguity. Among 240 primary queries with true effects above δ, ambiguity blocks 195 and sampling width 33; exact optima exclude optimization failure (Table III). Six additional finite certifications comprise four with sufficient standalone population information and two requiring fusion. Robust design retains the reference in 238/240 lower-budget cases: true effect zero.

Oracle radii use simulator truth. Across four radius/budget settings, fusion adds/loses 6/0, 9/0, 0/2 and 3/1 certifications versus the standalone union. Low-budget oracle metadata achieved gain averages 0.0114 pp versus 0.0913 pp budget cost; their difference changes the bound, not the policy effect. Proposition 1 proves zero optimal gain for 193/240 queries, including both losses; the larger-budget loss has positive gain below cost. All three losses persist at exact optima. Budget streams differ and robust policies can change across radii; fixed-world diagnoses estimate neither sample-size effects nor complementarity frequencies. Full diagnostics remain in the artifact.

a) Standalone-source controls: Observational consistency (4) lets OBS certify 99/100 external 2×2 queries; other policy/view settings differ. KuaiRec's imposed exposures and query-directed randomized means [24] do not establish general complementarity: a 35-setting census finds no true-radius fusion-only case. 400 count/policy pairs remain in the artifact.

The broader audit includes 6,000 inference views: 60 worlds, five repetitions, two R sample budgets, two radii and five constructions. The family's 2,880-view subset uses 48 main worlds and only OBS/projected-R/projected-Fusion. All 14 set misses occur outside that subset; none yields a bound above its true contrast or a false certification. All 18,900 final-mapping proofs (18,000 synthetic, 900 thinned Coat) pass independent checking; seven group-score certifications are unresolved after mapping, while projected statuses are unchanged. Separate design views have one miss in 600. Empirical coverage checks do not prove Theorem 1's assumptions.

TABLE IV
COAT: PRIMARY FIXED-QUERY LOWER BOUNDS (PROBABILITY UNITS; $\Lambda = 2$, $\rho = 0$, $\alpha = .05$; 290 USERS) AND FAMILY COUNTS SUMMING TO TWELVE DECLARATIONS. POSITIVE FAMILY ANSWERS OCCUR OUTSIDE THE PRIMARY DECLARATION. FULL-CELL TRANSPORT IS A STRONGER CONTROL.

Construction	Primary LCB	$\mathcal{R} = 0$	$0 < \mathcal{R} \le \delta$	$\mathcal{R} > \delta$
OBS	-0.03729	9/12	0/12	3/12
R group	-0.01502	12/12	0/12	0/12
R projected	-0.03402	12/12	0/12	0/12
Fusion group	-0.01193	7/12	1/12	4/12
Fusion projected	-0.03132	9/12	3/12	0/12
R full-cell control	-0.03077	1/1	0/1	0/1

D. Real data: inference over user blocks

Coat has 290 users, each supplying 24 self-selected and 16 randomized ratings on 300 items [25]. With one context ($K = 1$), policies sample items for this population, without personalization. Success is rating at least four, not purchase; zero is unobserved. Users, not the 11,600 ratings, are independent. Potential ratings are assumed invariant to within-source order and panel composition; cross-source equality needs an additional assumption.

The 31 moment groups partition items by gender, jacket type or color. The twelve declarations are $\Lambda \in \{1, 1.5, 2, 3\}$ and $\rho \in \{0, .02, .05\}$; the primary declaration is $(2, 0)$. Five binary indicators—front-page placement, men's items, fleece, waterproof and black—define fifteen policies $\pi_i \propto t^{h_i}$, $t \in \{.5, 1.5, 2\}$, relative to uniform $1/300$. The primary policy doubles front-page weights and renormalizes over all items. Only those three front-page contrasts lie outside the moment row space.

The main analysis keeps all ratings and assumes R16 simple random subsets independent of potential ratings. The release describes random/uniform selection; unavailable collection code prevents verification. A separate event-interface control samples one rating per source/user without using its value: $q_a = (16/300)/16 = 1/300$. Its 580 events leave 116 empty O and 124 empty R cells; all 900 queries and 15 full-cell controls are unresolved. This diagnoses the cost of discarding panel structure.

O conditional CP uses included users per item; joint CP uses $N = 290$. Uniform thinning divides marginal and latent inclusion propensities by 24, preserving their ratio and conditional ratings. Thus full inclusion strengthens consistency without changing Λ, not just reducing variance. Each active R user averages observed group ratings: under the subset design this is unbiased for their uniform group mean, with activity independent of potential ratings. Conditional on active count, these iid scores lie in $[0, 1]$. Equation (7) uses 72–290 users and variance denominator $n_j - 1$; item histograms cannot recover that variance. Projected intervals use per-item CP. Exact propagation follows (8), adding neither users nor transport evidence.

All 915 fixed-policy lower-bound queries are unresolved, with no refused view. At the primary declaration, group Fusion improves the bound over full-budget R group by 0.309 pp, but neither passes $\delta = .001$ (Table IV). Independent audit reconstructs user statistics, 61 views, 915 proofs and 15 direct controls; it does not establish the source assumptions.

Deciding beyond the tested policies. With the normalized ideal reference $1/300$, complete family optimization resolves all 60 original grid positions: 49 have $\mathcal{R} = 0$, four have $0 < \mathcal{R} \le .001$, and seven have $\mathcal{R} > .001$. The seven positive declarations represent five unique mathematical requests: OBS at $\Lambda = 1$ repeats across three tolerances; group Fusion passes at $\Lambda = 1$ for all three tolerances and at $\Lambda = 1.5, \rho = 0$. These supply checked policies although the original fifteen-candidate evaluation certified none, conditional on the stated assumptions; there is no joint guarantee across declarations.

A concrete policy decision. At $\Lambda = 1.5, \rho = 0$, the jacket-type certificate reallocates all 28 vests' uniform mass to item 0, a men's bomber: its probability rises from $1/300$ to $29/300$, the vests receive zero, and 271 items stay at $1/300$. All 15 original and 50 extended catalogue policies retain full support, so none makes this choice. The contrast is $(28/300)(m_0 - \overline{m}_{\text{vest}})$, with $\overline{m}_{\text{vest}}$ the uniform mean of the 28 vests. Its checked lower bound is $(28/300)(\ell_0 - b_{\text{vest}}/\kappa_{\text{vest}})$: the stored row mass κ converts the group constraint to its uniform mean. Only these two terms are nonzero, contributing 2.472638 pp and -2.226283 pp, respectively. Their sum guarantees 0.246355 pp in expected high-rating probability over uniform, exceeding 0.1 pp and matching the full overlapping view's capacity. Matched full-budget OBS and both R views have zero capacity. This retrospective answer assumes confounding no greater than $\Lambda = 1.5$ and no group-mean drift. It neither selects those assumptions from data nor measures purchase uplift; it does not imply every vest is worse.

All five primary $\Lambda = 2, \rho = 0$ views still have zero capacity. In group Fusion, every item mean can equal 0.2186 (rounded), tying every policy with the reference. A checked constant point already supplies this stopping answer; it is not an exclusive benefit of full optimization. The stronger full-cell control also has zero capacity.

The natural gender, jacket-type and color partitions supply 144 memberships: 10 exceed the threshold, 10 have positive capacity at most the threshold, and 124 have zero capacity. The construction and exact LP agree on all 95 unique supported requests across the full evaluation; the LP additionally resolves 30 overlapping requests. Every partition subview has capacity at most its parent's, as required by relaxation. These are dependent views of one dataset, not 144 independent experiments or new evidence for sampling assumptions.

Under full-cell transport, a separate direct control averages all 16 R ratings within each user and applies a one-sided Hoeffding bound ($N = 290$). The primary estimate is 0.00371, with lower bound -0.06452; none of 15 fixed queries passes. These per-query bounds neither support adaptive selection nor validate transport or a zero effect.

a) Catalogue queries with ordinary caching: For each primary Coat view, combine two of the five indicators: $\pi_i \propto$

TABLE V
ORIGINAL COAT QUERIES. TIME: MEDIAN SECONDS OVER THREE ROUNDS, INCLUDING STARTUP, FAMILY WORK, EXTERNAL VERIFICATION AND CLEANUP. RATIO/OPT: PAIRED COST RANGE (ABOVE ONE IS SLOWER). N/E/F: NATIVE/FIXED-EXACT/FAMILY CALLS, IDENTICAL PER STREAM ACROSS QUALITY AND TIMING. BOTH VIEWS USE $\rho = 0$.

View	Plan	Time (s)	Ratio/Opt	N/E/F
$\Lambda = 1.5$	Hot	6.800	0.974–1.009	2/2/0
$\Lambda = 1.5$	Opt	6.934	1.000	2/2/0
$\Lambda = 1.5$	Opt+Part	7.312	1.051–1.077	2/2/1
$\Lambda = 1.5$	Opt+LP	7.276	1.042–1.079	2/2/1
$\Lambda = 2$	Hot	6.329	0.985–0.998	1/1/0
$\Lambda = 2$	Opt	6.390	1.000	1/1/0
$\Lambda = 2$	Opt+Part	6.431	0.999–1.024	1/1/0
$\Lambda = 2$	Opt+LP	6.359	0.990–1.002	1/1/0

$t_1^{h_{1i}} t_2^{h_{2i}}$, with $(t_1, t_2) \in \{.5, 1, 2\}^2 \setminus \{(1, 1)\}$. Ten pairs give 80 policy positions at margins $0, .001, .01$ (0, 0.1, 1 pp): 240 requests, 50 distinct objectives and 150 objective–margin pairs. Prepared execution retains exact dual terms; optimized-exact adds sparse arithmetic and cached number conversions. Both share full objective caching and a persistent receiver. Native hot-start retains a solver basis without cross-objective certificate candidates. Quality checks precede three timing rounds.

Prepared and optimized-exact agree on all 1,200 requests, but cost ratios 0.960, 0.941, 0.953 miss the ten-percent target. Optimized-exact is slower in 3/15 view-rounds and retains more estimated exact-state payload in every view (not isolated RSS). Native hot-start changes 98 false answers to unknown and four unknown answers to false; it never reverses a Boolean answer. Thus overall service differs; on R projected, decisions agree and native is faster in all rounds. None certifies improvement, and neither fusion-only information target is supported. The five views have zero capacity, so a constant-action witness can already justify stopping.

b) The cost of family preprocessing: This experiment keeps the catalogue unchanged: it tests whether a family witness speeds up its existing queries, not whether to adopt the newly found policy. A useful policy outside the catalogue and false answers inside it can therefore coexist. We repeat the 240 queries on two 16-row jacket-type Fusion views, $(\Lambda, \rho) = (1.5, 0)$ and $(2, 0)$, selected retrospectively with known capacities. Hot-start (Hot), optimized threshold execution (Opt), and Opt with partition (Part) or exact-LP (LP) preprocessing share full objective caching, checked points, a constant shortcut and a rational receiver. Family certificates add only feasible points. The constant-action shortcut skips unnecessary family solves. Unresolved represented queries permit one producer attempt per objective–margin pair and one complete exact fallback per objective; ideal answers reuse this evidence without extra solves.

All 32 streams complete, each with 240 false answers under both semantics. On the nonzero-capacity view, Part adds 5.1–7.7% to Opt's full cost without reducing its two native and two exact calls (Table V); LP likewise saves no calls. The zero-capacity control skips family computation. Three rotations describe machine variation, with execution positions not fully balanced. Different optimal points compare complete preprocessing choices. The family answer enables redesign beyond this catalogue, without amortizing preprocessing cost here.

VII. RELATED WORK AND SCOPE

Our view combines binary density-ratio and conditional-moment bounds [1], [2], without latent-model sharpness. Related integration exploits overlap [26], fits responses under experimental moments [27], or estimates sensitivity frontiers [28]. Robust policy learning [29], [30] uses propensity-odds restrictions, unlike our outcome-conditional weights. Our increment is a direct partition construction and original-constraint certificate recovery within a common query interface.

Classical allocation. Continuous maximin sharing [31, Sec. 2.2] solves $\max_{0 \le z \le 1} \min_g \sum_{i \in G_g} v_i z_i$ under one resource budget, recovering capacities and allocations. Its groups partition profits; ours impose disjoint two-sided moments on means. We minimize $\max_i m_i - p^\top m$ per context and recover original-constraint certificates. The general exact LP supplies the same family answer and certificate contract; our measured speed advantage is limited to the smallest tested dimension.

Database queries. HypDB adjusts aggregates [32], ZaliQL implements matching [33], CaRL defines causal units [34], and CauSumX explains causal patterns [35]. Uncertainty annotations propagate bounds [36]; semantic views certify predicates from fresh draws [37]. Our materialized source-constrained means support new objectives under one coverage event and whole-family stopping certificates.

Proofs and limits. Filtering [38], assignment caching [39], safe bounds and reuse are established; VIPR checks exact optimization derivations [10]. Unlike IntegriDB [40], our digests authenticate neither data nor freshness. Binary outcomes, known target weights and trusted source/interval routines limit scope. Continuous covariates, estimated weights and dependent interactions need additional methods. Fine partitions can be uninformative; distant objectives favor reoptimization.

Data ethics. This secondary analysis uses public benchmarks under their release terms, without new recruitment or participant interaction. It underwent no institutional ethics review and sought no additional consent.

Code and material access: https://github.com/yzd2002/Q-BRIDGE-ICDE2027.

VIII. CONCLUSION

An analyst fixes a view, checks a constant-action stopping witness, then evaluates candidates or obtains a family certificate for redesign or exclusion. The partition construction certifies this decision against original constraints. Experiments support broader policy choice and exact-term retention in tested settings, with limits to fusion gains, screening and preprocessing.

Acknowledgments and AI Usage Declaration

OpenAI Codex assisted with drafting and mathematical exposition (Sections I–VIII), code and experiments (III–VI), figures, source retrieval and editing. Computational claims were checked against saved artifacts and citations against primary sources. The authors remain responsible for correctness, originality and the final submission.

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